Volume of prisms

1

a

i Triangle ii $12\,\text{cm}^2$ iii $276\,\text{cm}^3$

b

i Trapezium ii $40\,\text{cm}^2$ iii $240\,\text{cm}^3$

c

i Rectangle ii $24\,\text{m}^2$ iii $288\,\text{m}^3$

d

i Kite ii $16\,\text{cm}^2$ iii $32\,\text{cm}^3$

2

a $216\ \text{cm}^3$ b $48\ \text{cm}^3$ c $32\ \text{cm}^3$ d $217.5\ \text{cm}^3$

e $330\ \text{cm}^3$ f $84\ \text{cm}^3$ g $270\ \text{cm}^3$

3 $200\,\text{cm}^3$

4 $9600\,\text{mm}^3$

Volume of cylinders

5 $V=\pi r^2 h$

6

a Circle b $36\pi\ \text{cm}^2$ c $360\pi\ \text{cm}^3$

7

a $10\,995.6\ \text{units}^3$ b $2\,413\,754.1\ \text{units}^3$ c $785.4\ \text{cm}^3$ d $240.3\ \text{cm}^3$

e $45.9\ \text{cm}^3$ f $70.7\ \text{cm}^3$ g $307.9\ \text{cm}^3$ h $1608.5\ \text{units}^3$

i $883.6\ \text{cm}^3$ j $263.9\ \text{cm}^3$ k $609.5\ \text{cm}^3$

Volume of compound solids

8

a

i A semicircle and a square ii $22.283\,\text{m}^2$

iii $222.83\,\text{m}^3$

b

i Two rectangles

ii $34\,\text{cm}^2$

iii $68\,\text{cm}^3$

c

i Two rectangles

ii $40\,\text{cm}^2$

iii $280\,\text{cm}^3$

d

i A triangle and a rectangle

ii $37.5\,\text{m}^2$

iii $412.5\,\text{m}^3$

9

a 399 cm^3

b $1080\,\text{cm}^3$

c 828 cm^3

d $982.35\,\text{m}^3$

e 300 cm^3

f 293.52 cm^3

Unknown side given the volume

10 $a = 4\text{ mm}$

11 $p = 6\text{ mm}$

12 $k = 11$

13 $y = 10.1$

14 $70\,\text{cm}^2$

15 $9\,\text{cm}$

16 3 cm

17 9 cm

Volume of spheres

18

a 113.10 cm^3

b 4577.20 cm^3

c $113\,550.33\text{ mm}^3$

d 33.51 cm^3

19 The student's mistake was substituting the [redacted]eter for the radius. Correct volume is 381.70 cm^3

20

a 817.283 mm^3 b 2572.441 cm^3 c 3.054 m^2 d 0.351 cm^2

Applications

21 2 m^3

22

a 29.69 cm^3 b 169.2 g

23 0.57 m^3

24

a 1335.18 cm^3 b 4005.54 cm^3 c 28 L

25

a 1198.5 m^3 b 299.625 m^3 c $h = 0.5875 \text{ m}$

26

a 8151 cm^3 b 5

27

a $2 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$ b 20 cm^3

c $15\,120 \text{ cm}^3$ d 756

28 $d = 29 \text{ cm}$

29 $h = 160 \text{ cm}$

30 5424.605 cm^3

31 8000 balls

32 $113\,097 \text{ cm}^3$

34 1.646×10^{11} km^3

35

a 32.5542 cm

b 18 064 cm^3

36

a 804.2 cm^3

b 1206.4 cm^3

c 33%